Ch. 11 CW

Principles of Microeconomics

July 23, 2026

Instructions

Please justify all of your answers and submit your solutions on Gradescope by the beginning of next class

Problems

1. Four roommates are planning to spend the weekend in their dorm room watching old movies, and they are debating how many to watch. Here is their willingness to pay for each film:

	Dwayne	Javier	Salman	Chris
First film	$7	$5	$3	$2
Second film	6	4	2	1
Third film	5	3	1	0
Fourth film	4	2	0	0
Fifth film	3	1	0	0

   (a) Within the dorm room, is the showing of a movie a public good? Why or why not?

   (b) If it costs $8 to stream a movie, how many movies should the roommates stream to maximize total surplus?

   (c) If they choose the optimal number from part (b) and then split the cost of streaming the movies equally, how much surplus does each person obtain from watching the movies?

   (d) Is there any way to split the cost to ensure that everyone benefits? What practical problems does this solution raise?

   (e) Suppose they agree in advance to choose the efficient number and to split the cost of the movies equally. When Dwayne is asked his willingness to pay, will he have an incentive to tell the truth? If so, why? If not, what will he be tempted to say?

   (f) What does this example teach you about the optimal provision of public goods?

2. Some economists argue that private firms will not undertake the efficient amount of basic scientific research.

   (a) Explain why this might be so. In your answer, classify basic research as one of the four types of goods.

   (b) What sort of policy has the United States adopted in response to this problem?

   (c) It is often argued that this policy increases the technological capability of American producers relative to that of foreign firms. Is this argument consistent with your classification of basic research in part (a)? (Hint: Can excludability apply to some potential beneficiaries of a public good and not others?)

3. Two towns, each with three residents, are deciding whether to put on a fireworks display to celebrate the New Year. Fireworks cost $360. In each town, some people enjoy fireworks more than the others.

   (a) In the town of Bayport, each of the residents values the public good as follows:

Frank	$50
Joe	$100
Callie	$300

   Would fireworks pass a cost-benefit analysis? Explain.

   (b) The mayor of Bayport proposes to decide by majority rule and, if the fireworks referendum passes, to split the cost equally among all residents. Who would vote in favor, and who would vote against? Would the vote yield the same answer as the cost-benefit analysis?

   (c) In the town of River Heights, each of the residents values the public good as follows:

Nancy	$20
Bess	$140
Ned	$160

   Would fireworks pass a cost-benefit analysis? Explain.

   (d) The mayor of River Heights also proposes to decide by majority rule and, if the fireworks referendum passes, to split the cost equally among all residents. Who would vote in favor, and who would vote against? Would the vote yield the same answer as the cost-benefit analysis?

   (e) What do you think these examples say about the optimal provision of public goods?

4. Many transportation systems, such as the Washington, D.C., Metro (subway), charge higher fares during rush hours than during the rest of the day. Why might they do this?

5. Four roommates are thinking about hiring someone to clean their apartment. The cost of cleaning the apartment is \$20. One roommate's demand for cleanings is given by $Q_D = 5 - \frac{1}{2}P$. The other three roommates each have the same individual demand equation for cleanings, $Q_D = 3 - \frac{1}{2}P$.

   (a) What is the market demand for cleanings amongst the four roommates?

   (b) What is the optimal number of cleanings they should buy?

   (c) If the roommates didn't get along and couldn't work together long enough to hire a cleaner, would any of them individually pay for cleanings?